
Lab 03 Report

Cascode Amplifier

1. Part 1: Sizing Chart

Intrinsic Gain and V^* Definition

The intrinsic gain of a MOSFET is:

$$|A_v| \approx g_m r_o = \frac{2I_D}{V_{ov}} \cdot \frac{V_A}{I_D} = \frac{2V_A}{V_{ov}}$$

For a real MOSFET the square-law overestimates g_m . We define:

$$V^* = \frac{2I_D}{g_m} \quad \Leftrightarrow \quad g_m = \frac{2I_D}{V^*} \quad \Rightarrow \quad |A_v| \approx \frac{2V_A}{V^*}$$

A commonly used sweet-spot balancing gain, area, and speed is $V^* = 200 \text{ mV}$.

Design Parameters

Table 1: Design Specifications

Parameter	0.18 μm CMOS
V^*	200 mV
Supply V_{DD}	1.8 V
Current I_{DQ}	30 μA
Intrinsic gain $g_m r_o$	90 V/V

Characterisation Testbench

Create an NMOS testbench with $W = 10 \mu\text{m}$ (reference), and sweep L from 180 nm to $2 \mu\text{m}$ since the requirement of intrinsic gain is small, A short channel device will be the optimum choice.

V^*, V_{ov} , and $\frac{g_m}{g_{ds}}$ vs V_{GS} with L parameterized

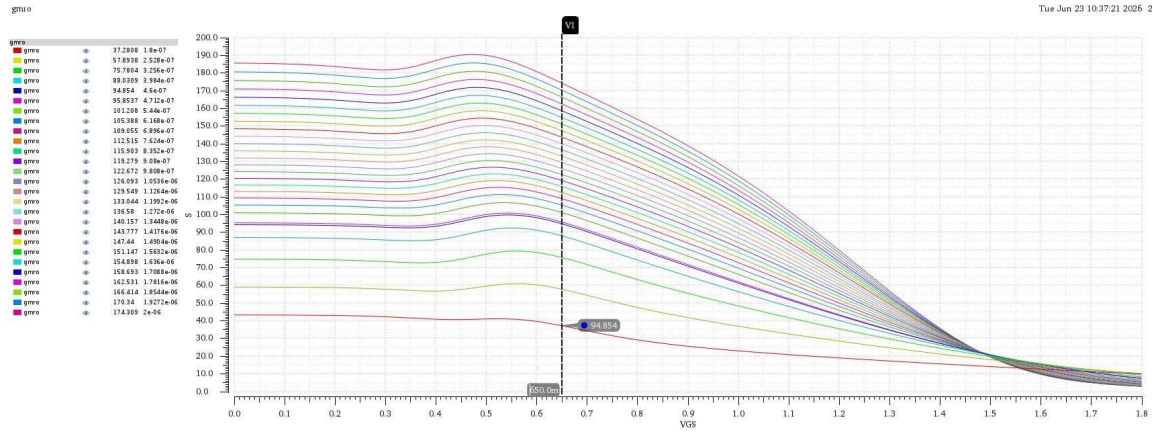


Figure 1: g_m/g_{ds} vs V_{GS} at various L

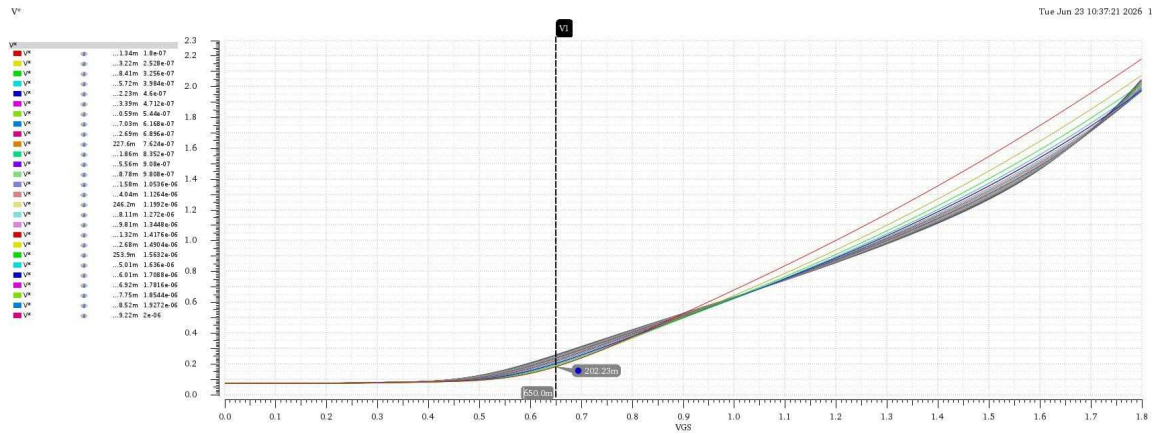
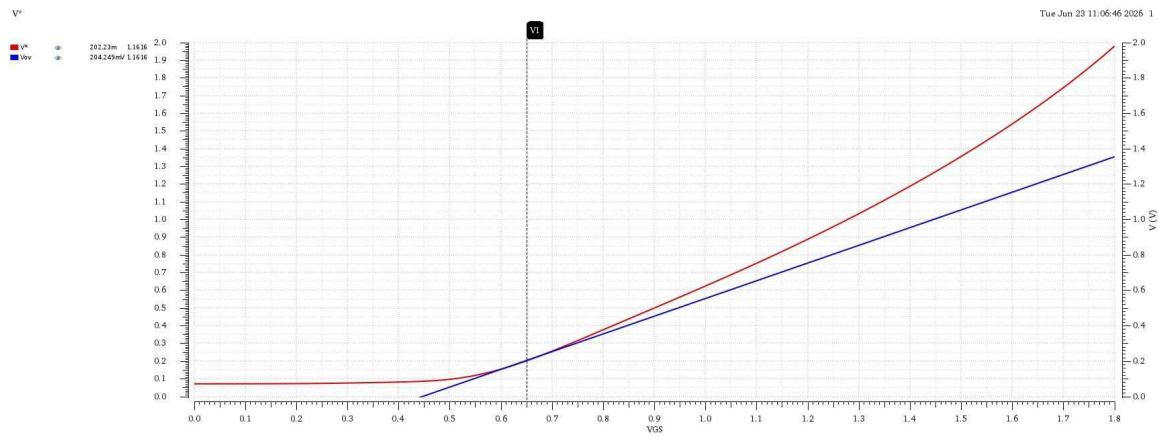
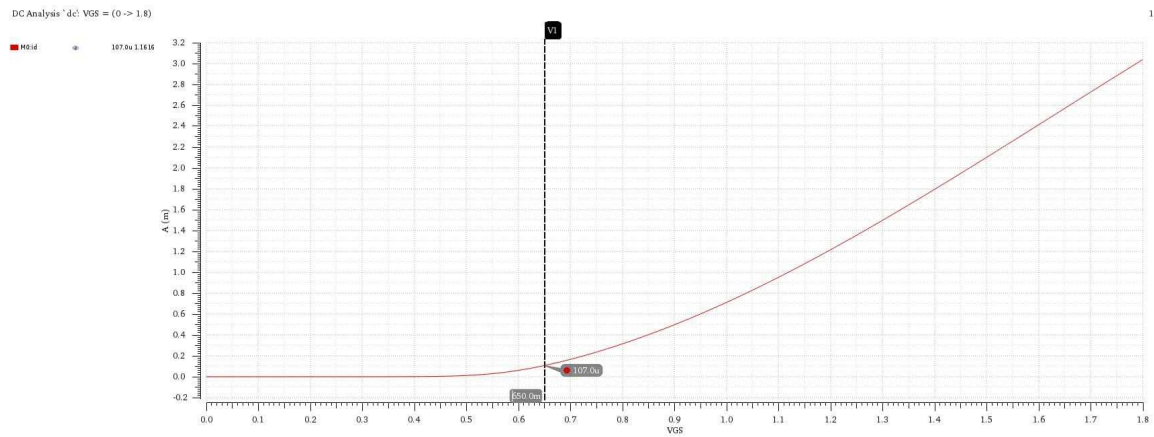
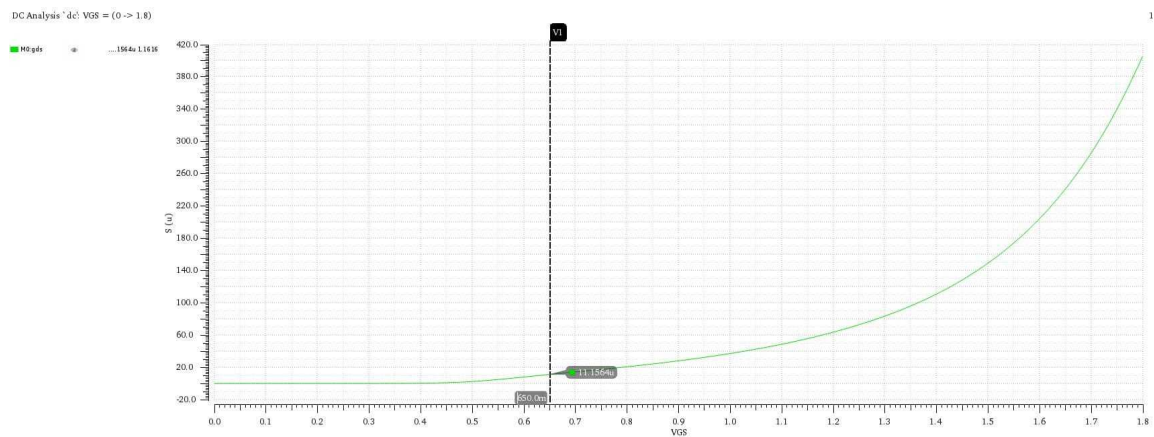


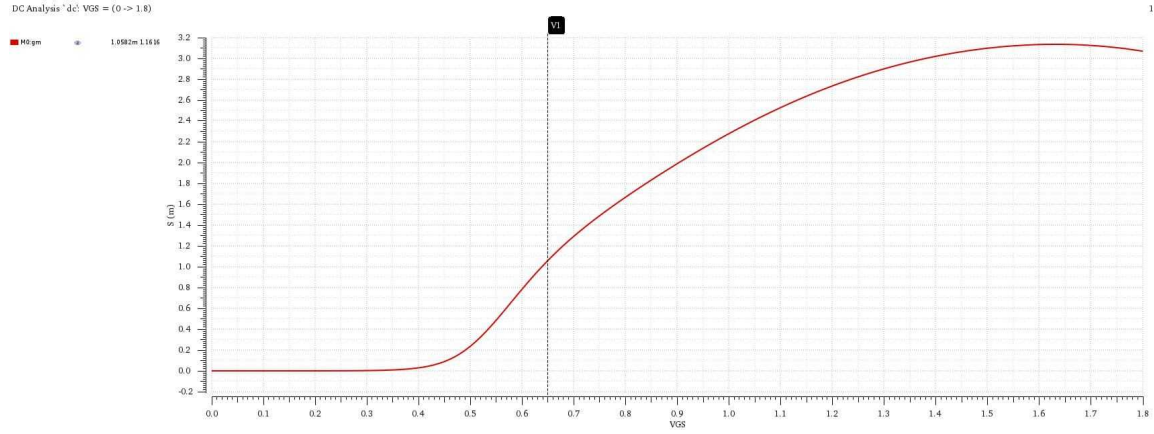
Figure 2: V^* vs V_{GS} at various L

From the graphs, At $V_{GS} = 650 mV$ and $L = 460 nm$, The intrinsic gain is 94.854 and $V^* = 202.23 mV$.

Figure 3: V^* and V_{ov} vs V_{GS} at $L = 460nm$

I_D , g_m , g_{ds} vs V_{GS}

Figure 4: I_D vs V_{GS} at $L = 460nm$ Figure 5: g_{ds} vs V_{GS} at $L = 460nm$

Figure 6: g_m vs V_{GS} at $L = 460nm$ Table 2: Reference-Width Parameters at $V_{GSQ} = 650mV$

Parameter	0.18 μm
I_{DX} (μA)	107
g_{mX} (mA/V)	1.0582
g_{dsX} (μS)	11.1564

Scale Width to $I_{DQ} = 30 \mu A$

$$\frac{W}{10 \mu m} = \frac{I_{DQ}}{I_{DX}} \Rightarrow W = \frac{30 \mu A}{I_{DX}} \times 10 \mu m$$

Table 3: Width Scaling (Cross-Multiplication)

W	I_D
10 μm	107 μA
$W = 2.803738 \mu m$	$I_{DQ} = 30 \mu A$

Scale g_{mQ} and g_{dsQ}

$$g_{mQ} = g_{mX} \times \frac{W}{10 \mu m}, \quad g_{dsQ} = g_{dsX} \times \frac{W}{10 \mu m}, \quad r_o = \frac{1}{g_{dsQ}}$$

Calculation of g_{mQ} , g_{dsQ} , r_o for 0.18 μm CMOS

$$g_{mQ} = g_{mX} \times \frac{W}{10 \mu\text{m}} = 1.0582 \text{mA/V} \times \frac{2.803738}{10} = 296.6915 \mu\text{A/V}$$

$$g_{dsQ} = g_{dsX} \times \frac{W}{10 \mu\text{m}} = 11.1564 \mu\text{S} \times \frac{2.803738}{10} = 3.128 \mu\text{S}$$

$$r_o = \frac{1}{g_{dsQ}} = 319.696 \text{k}\Omega$$

$$g_m r_o = 296.6915 \mu\text{A/V} * 319.696 \text{k}\Omega = 94.8513 \text{V/V}$$

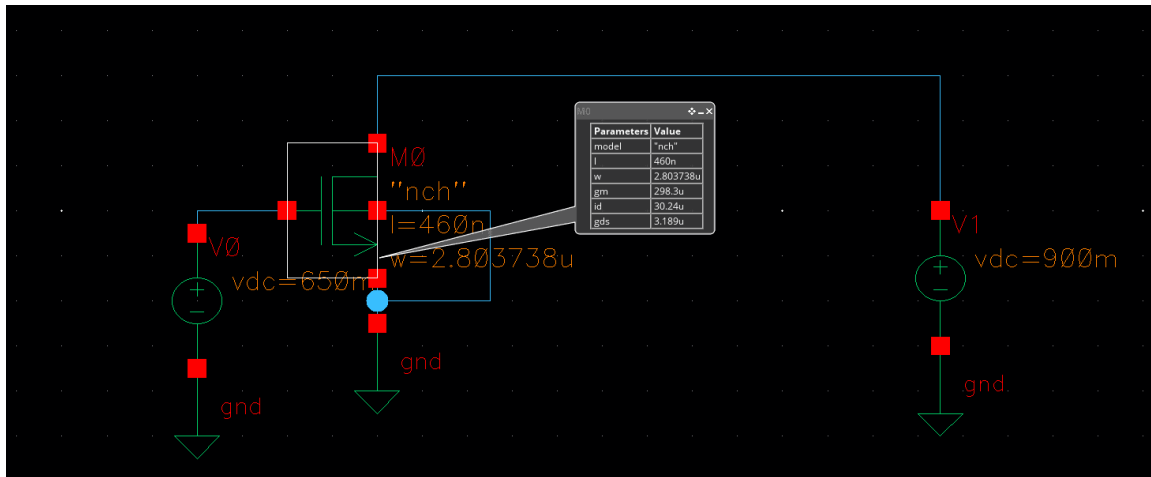


Figure 7: Schematic DC Operating Points

2. Part 2: Cascode for Gain

2.1 Choose V_B so that $V_{DS0} = V^* + 100\text{mV}$

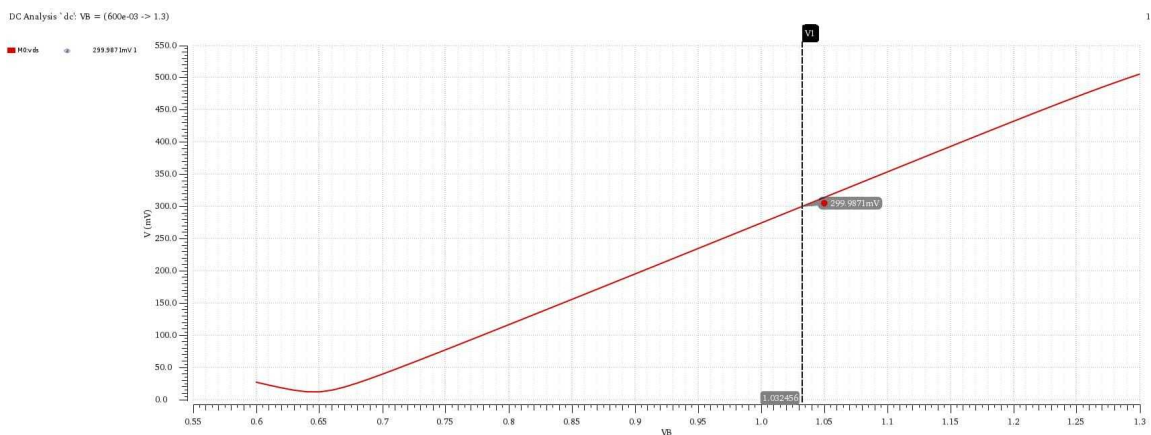


Figure 8: V_{DS} vs V_B

DC Operating Point Simulation

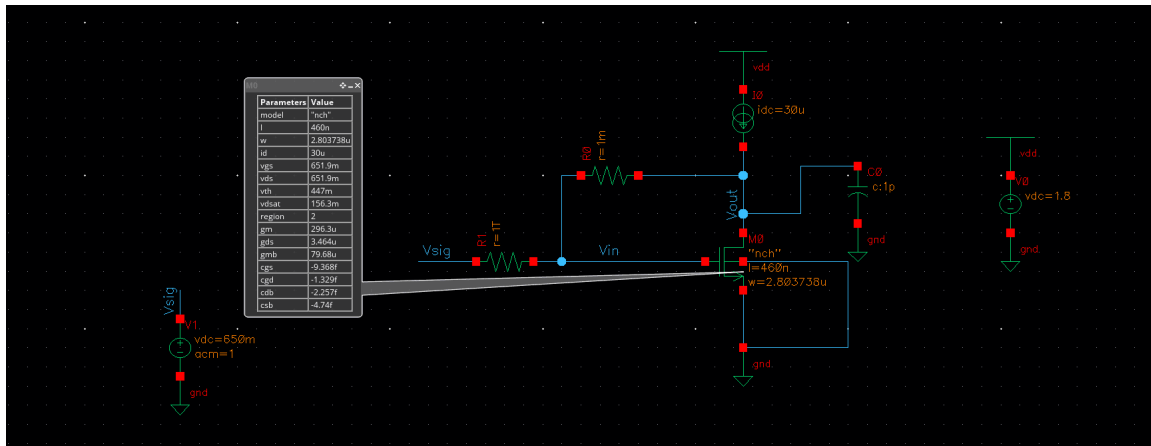


Figure 9: Single Stage CS Amplifier

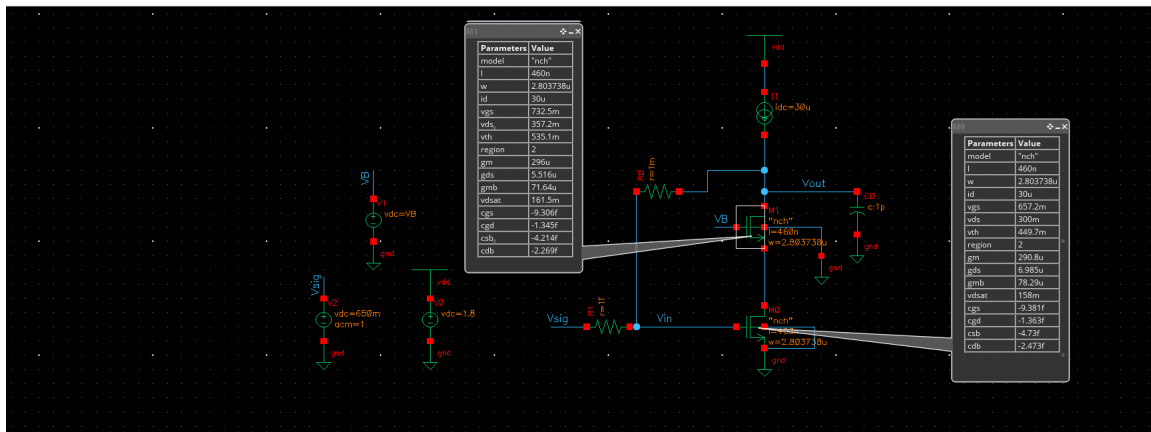


Figure 10: Cascode for Gain Schematic

Saturation Check

Yes, All the transistors are in saturation.

Threshold Voltage V_{TH} Comparison

Not all the transistors have the same v_{th} . $M0$ in both schematics have nearly the same v_{th} as their sources are tied to ground ($V_{sb} = 0$). However, $M1$ suffers from body effect as source is not tied to ground leading to higher v_{th}

Parameter Relations

Table 4: Small-Signal Parameter Relations

Relation
$g_m \gg g_{ds}$
$g_m > g_{mb}$
$C_{gs} > C_{gd}$
$C_{sb} > C_{db}$

2.2 AC Analysis

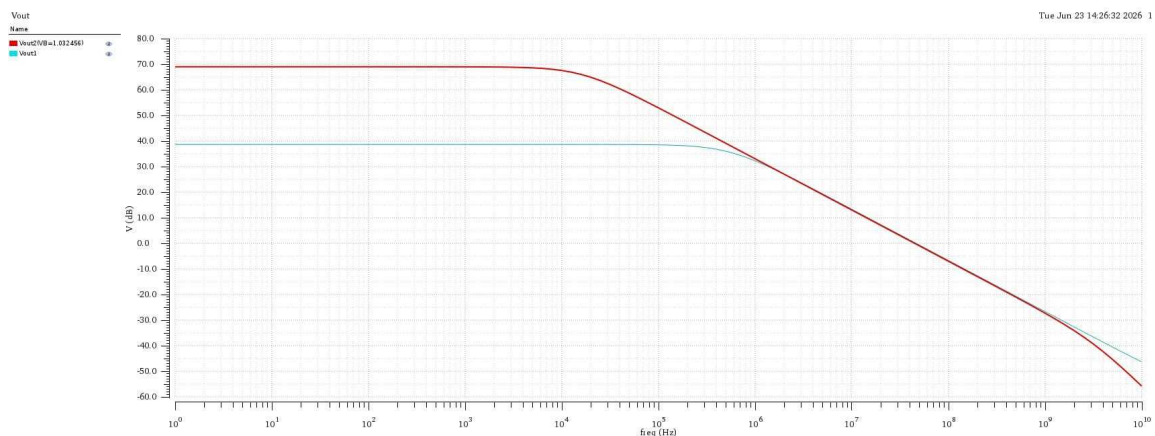


Figure 11: Gain Comparison (Vout1- $\gg$ Single, Vout2- $\gg$ Cascode)

Test	Output	Nominal
Cascode:Single:1	Vout1	
Cascode:Single:1	y _{max} (dB20(VF("/Vout")))	38.64
Cascode:Single:1	y _{max} (mag(VF("/Vout")))	85.53
Cascode:Single:1	BWSingle	549.8k
Cascode:Single:1	GBWsingle	47.14M
Cascode:Single:1	UGF	47.46M

Figure 12: Single stage AC Results.

Test	Output	Nominal
Cascode:Cascode:1	Vout2	
Cascode:Cascode:1	y _{max} (dB20(VF("/Vout")))	68.99
Cascode:Cascode:1	y _{max} (mag(VF("/Vout")))	2.816k
Cascode:Cascode:1	BW2	16.04k
Cascode:Cascode:1	GBW2	45.29M
Cascode:Cascode:1	UGF2	45.83M

Figure 13: Cascode for Gain AC Results.

Hand Analysis

CS Amplifier (VOUT1)

DC Gain, BW, and GBW — CS Amplifier (hand analysis)

$$A_{v,CS} = -g_{m1}(r_{o1} \parallel r_{o3}) = -296.3 \mu\text{S} \times (288.683 \text{ k}\Omega \parallel \infty) = -85.536 \text{ V/V}$$

$$\omega_p \approx \frac{1}{(r_{o1} \parallel r_{o3}) C_L} \Rightarrow f_{-3\text{dB}} = \frac{1}{2\pi \times 288.683 \text{ k}\Omega \times 1 \text{ pF}} = 551.313 \text{ kHz}$$

$$\text{GBW} = |A_{v,CS}| \times f_{-3\text{dB}} = 85.536 \times 551.313 \text{ kHz} = 47.157 \text{ MHz}$$

Cascode Amplifier (VOUT2)**DC Gain, BW, and GBW — Cascode Amplifier (hand analysis)**

Output resistance of cascode stage:

$$R_{out,casc} \approx (r_{o1} + r_{o2} + (g_{m2} + g_{mb2})r_{o2}r_{o1}) \parallel \infty = 9.866 \text{ M}\Omega \parallel \infty = 9.866 \text{ M}\Omega$$

$$A_{v,casc} = -g_{m1}R_{out,casc} = -290.8 \mu\text{S} \times 8.01 \text{ M}\Omega = -2869.118 \text{ V/V}$$

$$f_{-3\text{dB},casc} = \frac{1}{2\pi R_{out,casc} C_L} = \frac{1}{2\pi \times 9.866 \text{ M}\Omega \times 1 \text{ pF}} = 16.131 \text{ kHz}$$

$$\text{GBW}_{casc} = |A_{v,casc}| \times f_{-3\text{dB},casc} = 2869.118 \times 16.131 \text{ kHz} = 46.28 \text{ MHz}$$

Comparison Table

Table 5: CS vs Cascode — Gain, BW, GBW, UGF

Parameter	CS Amplifier		Cascode Amplifier	
	Simulation	Hand Analysis	Simulation	Hand Analysis
DC Gain (V/V)	85.53	85.536	2816	2869.118
DC Gain (dB)	38.64	38.643	68.99	69.155
BW (kHz)	549.8	551.313	16.04	16.131
GBW (MHz)	47.14	47.157	45.29	46.28
UGF (MHz)	47.46	47.157	45.83	46.28

Comments

The cascode amplifier achieves a significantly higher DC gain than the common-source (CS) amplifier by boosting the output resistance from roughly r_o to $g_m r_o^2$. However, because the dominant pole is located at the output node ($f_{-3dB} = \frac{1}{2\pi R_{out} C_L}$), this massive increase in output resistance causes a proportional, drastic reduction in the bandwidth.

Despite this extreme trade-off between gain and bandwidth, the GBW remains virtually identical for both topologies. This is because the GBW is fundamentally dictated by the transconductance of the input device and the load capacitance ($GBW \approx \frac{g_m}{2\pi C_L}$). Since both circuits use the exact same input transistor (g_{m1}) biased at the same current, and drive the same load capacitance (C_L), their frequency capability is constant. Ultimately, cascoding simply shifts the dominant pole to trade bandwidth for DC gain.

3. Part 3 [Optional]: Cascode for BW

3.1 OP Analysis

Modified Schematic

Keeping $V_{DS} = 900mV$

$$R_D = \frac{V_{DS}}{30\mu A} = 30k\Omega$$

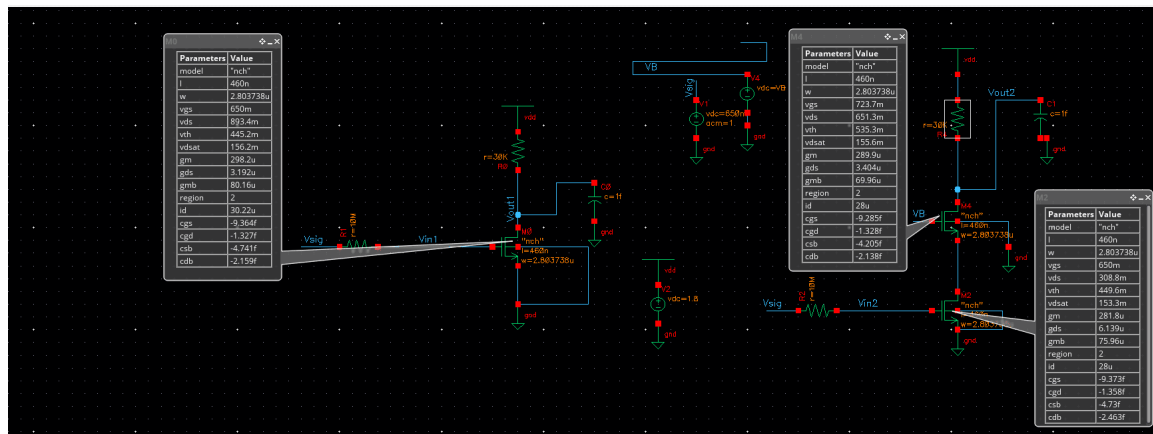


Figure 14: Cascode for BW Schematic with DC Operating point

Saturation Check

Yes, All the transistors are in saturation.

3.2 AC Analysis

Bode Plot

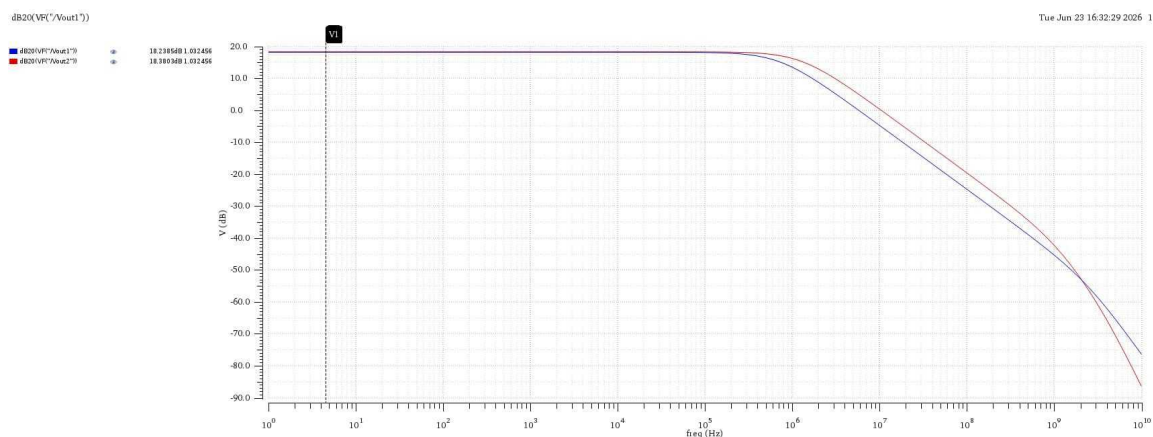


Figure 15: Magnitude Bode plot for Vout1≫Single, Vout2≫Cascode

Test	Output	Nominal
BW:BW:1	dB20(VF("/Vout1"))	↙
BW:BW:1	ymax(dB20(VF("/Vout1")))	18.24
BW:BW:1	ymax(mag(VF("/Vout1")))	8.164
BW:BW:1	bandwidth(VF("/Vout1") 3 "lo...	721.3k
BW:BW:1	gainBwProd(VF("/Vout1"))	5.903M
BW:BW:1	dB20(VF("/Vout2"))	↙
BW:BW:1	ymax(dB20(VF("/Vout2")))	18.38
BW:BW:1	ymax(mag(VF("/Vout2")))	8.299
BW:BW:1	bandwidth(VF("/Vout2") 3 "lo...	1.281M
BW:BW:1	gainBwProd(VF("/Vout2"))	10.66M
BW:BW:1	unityGainFreq(VF("/Vout2"))	10.67M
BW:BW:1	unityGainFreq(VF("/Vout1"))	5.905M

Figure 16: AC results

4 — Hand Analysis

CS Amplifier with R_D

DC Gain, BW, GBW — CS with R_D

$$A_{v,CS} = -g_{m1}(R_D \parallel r_{o1}) = -298.2 \mu\text{S} \times (30 \text{ k}\Omega \parallel 313.28 \text{ k}\Omega) = -8.165 \text{ V/V}$$

Note: With $R_{sig} = 10 \text{ M}\Omega$, the input pole dominates due to the Miller Effect.

$$C_{in} = C_{gs} + C_{gd}(1 + |A_{v,CS}|) = 9.364 \text{ fF} + 1.327 \text{ fF}(1 + 8.165) = 21.52 \text{ fF}$$

$$f_{-3\text{dB}} \approx \frac{1}{2\pi R_{sig} C_{in}} = \frac{1}{2\pi \times 10 \text{ M}\Omega \times 21.52 \text{ fF}} = 739.6 \text{ kHz}$$

$$\text{GBW} = |A_{v,CS}| \times f_{-3\text{dB}} = 8.165 \times 739.6 \text{ kHz} = 6.039 \text{ MHz}$$

Cascode Amplifier with R_D

DC Gain, BW, GBW — Cascode with R_D

$$R_{out,casc} \approx \left((g_{m2} + g_{mb2})r_{o2}r_{o1} \right) \parallel R_D \approx 17.68 \text{ M}\Omega \parallel 30 \text{ k}\Omega = 29.95 \text{ k}\Omega$$

$$A_{v,casc} = -g_{m1}R_{out,casc} = -281.8 \mu\text{S} \times 29.95 \text{ k}\Omega = -8.44 \text{ V/V}$$

Because the intermediate node gain $A_{v1} \approx -1$, the Miller multiplication is drastically reduced:

$$C_{in,casc} \approx C_{gs1} + C_{gd1}(1 + |-1|) = 9.373 \text{ fF} + 1.358 \text{ fF}(2) = 12.09 \text{ fF}$$

$$f_{-3\text{dB},casc} \approx \frac{1}{2\pi R_{sig} C_{in,casc}} = \frac{1}{2\pi \times 10 \text{ M}\Omega \times 12.09 \text{ fF}} = 1.316 \text{ MHz}$$

$$\text{GBW}_{casc} = |A_{v,casc}| \times f_{-3\text{dB},casc} = 8.44 \times 1.316 \text{ MHz} = 11.11 \text{ MHz}$$

Comparison Table and Comments

Table 6: CS vs Cascode with R_D — Gain, BW, GBW, UGF

Parameter	CS with R_D		Cascode with R_D	
	Simulation	Hand Analysis	Simulation	Hand Analysis
DC Gain (V/V)	8.164	8.165	8.299	8.44
DC Gain (dB)	18.24	18.24	18.38	18.53
BW (MHz)	0.721	0.740	1.281	1.316
GBW (MHz)	5.903	6.039	10.66	11.11
UGF (MHz)	5.905	6.039	10.67	11.11

In Part 2, the cascode topology traded a massive reduction in bandwidth for a drastically higher DC gain, keeping the overall Gain-Bandwidth Product (GBW) identical to the CS amplifier. However, introducing a resistive load (R_D) in Part 3 caps the maximum DC gain for both configurations. With the gain restricted, the cascode's intermediate node gain drops ($A_{v1} \approx -1$), effectively neutralizing the Miller effect on C_{gd} . Consequently, the cascode achieves a significantly higher bandwidth and a much larger overall GBW than the CS amplifier, fundamentally breaking the traditional constant-GBW trade-off.